ShadeSpot — Frontend Architecture & Product Context

> Version 1.0 · April 2026
> Production reference document for frontend engineering, design, and product.
> All decisions in this document reflect the dual-pool FHE privacy architecture and the selective disclosure narrative.

Table of Contents

1. Product Philosophy
2. User Segments & Mental Models
3. Application Structure & Routing
4. Design System
5. Wallet & Authentication Layer
6. Pool Selection & Onboarding
7. Trade Page — Pool 1 (USDC)
8. Trade Page — Pool 2 (FHE Token)
9. Position Management & Decryption UX
10. Order Management
11. Liquidation UX
12. Earn Page — Liquidity Provision
13. Analytics Page
14. Governance Page
15. Privacy Centre
16. Notification & Alert System
17. State Management Architecture
18. CoFHE SDK Integration
19. Performance & Loading States
20. Error Handling & Edge Cases
21. Responsive & Mobile
22. Accessibility
23. Security Considerations
24. Environment & Deployment

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1. Product Philosophy

1.1 The Core Tension

ShadeSpot exists in direct opposition to the default transparency of public blockchains. Every design decision must resolve the same tension:

> Privacy should be invisible when it is protecting you, and legible when you choose to look at it.

This means the trading experience must feel as fast and clean as Hyperliquid or GMX for a casual user, while surfacing the cryptographic guarantees clearly for a user who cares about them. These are not contradictory — they require layered disclosure.

1.2 Three Design Mandates

Mandate 1 — Blur by default, reveal by choice.
No position data is shown unless the user has explicitly decrypted it. Blurred values with a lock icon are the default state for every sensitive field. The act of decryption is a deliberate, visible gesture — not something that happens automatically.

Mandate 2 — Label the privacy guarantee on every number.
Every aggregate statistic on the platform carries a badge indicating its privacy model: `ZK` for zero-knowledge proven exact values, `DP` for differentially private approximate values, or `PUBLIC` for plaintext on-chain data. Users should never have to wonder whether a number is real or estimated.

Mandate 3 — Never apologise for latency.
Threshold Network decryption takes 1–2 seconds. Do not show a spinner and hope the user waits. Use skeleton loaders, progress indicators with copy ("Requesting decryption from Threshold Network…"), and immediate confirmation of success. Make the decryption round-trip feel intentional and trustworthy, not slow.

1.3 What We Are Not Building

- We are not building a Bloomberg terminal with 40 panels.
- We are not building a consumer crypto app that hides complexity entirely.
- We are not building a dashboard that looks impressive but is hard to act from.

We are building a professional trading interface for people who understand leverage and have strong opinions about privacy.

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2. User Segments & Mental Models

2.1 Segment A — Privacy-Conscious Retail Trader

Profile: Experienced DeFi trader who has been front-run or liquidation-hunted on Hyperliquid or

GMX. Understands leverage mechanics well. Moderately technical — understands wallets, gas, and transactions but not cryptography.

Mental model of privacy: "My positions are hidden like they would be in a CEX dark pool. Nobody can see what I'm doing."

Key needs:

- Fast order entry, feels like a CEX
- Clear confirmation that their position is encrypted (one visible signal per action is enough)
- Easy decryption of their own data when they want to review it
- Pool 1 (USDC) is their entry point — familiar collateral

Friction points to eliminate:

- Having to understand what FHE is before placing a trade
- Multiple steps to see their own position data
- Uncertainty about whether the privacy is real or marketing

2.2 Segment B — Institutional / Sophisticated Trader

Profile: Portfolio manager, prop desk trader, or sophisticated individual with positions above $50,000. Has compliance awareness. Cares about proving the integrity of the system to themselves and potentially to others.

Mental model of privacy: "My strategy and position sizing are confidential. I can selectively prove my positions to an auditor without revealing them to the market."

Key needs:

- Selective disclosure — ability to share a permit with a specific address (auditor, counterparty)
- Cryptographic proof that privacy is structural, not a promise
- Pool 2 (FHE Token) if they want fully encrypted LP participation
- Access to ZK-proven aggregate analytics for market context

Friction points to eliminate:

- Needing to trust the team rather than the math
- Lack of audit trail for their own historical positions
- Uncertainty about which analytics numbers are real vs approximate

2.3 Segment C — Liquidation Bot Operator

Profile: Developer running keeper or liquidation infrastructure. Cares about efficiency, reliability, and bot economics.

Mental model of privacy: "I need to know which positions are flagged. I don't need to know the position details."

Key needs:

- Clean public API for querying flagged positions and the liquidation queue
- Price bucket heatmap to estimate where to probe
- Gas cost transparency (FHE checks cost more — bots need to know)
- Webhook or event subscription for PositionFlagged events

Not a UI user — primarily served by documentation, the analytics API, and the on-chain event stream.

2.4 Segment D — Liquidity Provider

Profile: Yield-seeking capital allocator. May use Pool 1 (plaintext LP shares) or Pool 2 (encrypted LP shares). Understands that LPs are the counterparty to all trades.

Mental model of privacy: Pool 1 — "I can see my balance and the vault utilisation." Pool 2 — "My LP position is private. I decrypt it when I want to see it."

Key needs:

- Clear vault utilisation and projected yield
- Pool 2 — same decrypt-by-permit pattern as traders
- Understanding of LP risk (they are the counterparty)
- Withdrawal mechanics, including the check against available liquidity

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3. Application Structure & Routing

3.1 Top-Level Routes

```
/                       → Redirect to /trade
/trade                  → Trade page (defaults to Pool 1, BTC-PERP)
/trade/:pool/:market    → e.g. /trade/1/BTC-PERP or /trade/2/ETH-PERP
/earn                   → Liquidity provision page
/earn/:pool             → /earn/1 or /earn/2
/analytics              → Public analytics dashboard
/analytics/:market      → Per-market analytics
/govern                 → Governance page
/privacy                → Privacy Centre (explains the system)
/positions              → My positions across both pools (requires wallet)
/history                → Trade history (requires wallet + optional decryption)
```

```
/settings                → User preferences, permit management, notification
settings
```

3.2 Layout Shell

The application uses a persistent shell with three regions:

```
┌──────────────────────────────────────────────────────────────┐
│  NAVBAR                                                      │
│  Logo | Nav links | FHE status pill | Network | Wallet       │
├──────────────────────────────────────────────────────────────┤
│  CONTEXT BAR (page-specific)                                 │
│  Trade: ticker strip                                         │
│  Analytics: market selector                                  │
│  Earn: pool selector + vault stats                           │
├──────────────────────────────────────────────────────────────┤
│  MAIN CONTENT (page-specific)                                │
│                                                              │
├──────────────────────────────────────────────────────────────┤
│  FOOTER (minimal)                                            │
│  Docs | Github | Audit | Privacy policy | Status             │
└──────────────────────────────────────────────────────────────┘
```

3.3 Persistent Global State Requirements

The following must be available across all pages without re-fetching:

- Connected wallet address and chain
- Active pool (1 or 2)
- Oracle prices (subscribed via WebSocket)
- User's open positions (encrypted handles, decrypted lazily)
- Active permit status per position
- Notification queue

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4. Design System

4.1 Visual Identity

ShadeSpot's visual language is **dark terminal meets institutional trading desk**. It should feel like something a prop trader would use — dense, information-rich, precise — not a retail DeFi app.

Reference aesthetic: Bloomberg terminal darkness + Hyperliquid precision + the monospaced clarity of a code editor.

Fonts:

- Primary: `Geist Mono` — used for all numerical values, addresses, prices, percentages, and any data that originates from the chain. Monospaced numbers prevent layout shift as values update.
- Secondary: `Geist` — used for labels, descriptions, navigation, and copy. Clean sans-serif that pairs with the mono without clashing.

Rationale for Geist family: Both fonts are from Vercel's open-source type system. They are designed to coexist. The mono/sans distinction creates a visual grammar: mono = data from chain, sans = human annotation.

4.2 Colour Palette

```
css
```

```
/* Backgrounds */
--bg-primary:     #07090d;   /* Page background */
--bg-surface:     #0c0f15;   /* Cards, panels, navbar */
--bg-elevated:    #111620;   /* Inputs, dropdowns, hover states */
--bg-high:        #16202e;   /* Active states, selected rows */

/* Borders */
--border-subtle:  rgba(255,255,255,0.05);
--border-default: rgba(255,255,255,0.08);
--border-strong:  rgba(255,255,255,0.12);

/* Text */
--text-primary:   #e8edf5;   /* Main content */
--text-secondary: #7a8fa8;   /* Labels, captions */
--text-muted:     #3d5068;   /* Disabled, placeholder */

/* Brand */
--teal:           #06d6a0;   /* Primary brand — FHE encrypted indicator */
--teal-dim:       rgba(6,214,160,0.08);
--teal-border:    rgba(6,214,160,0.18);

/* Semantic */
--green:          #16c784;   /* Profit, long, positive */
--green-dim:      rgba(22,199,132,0.10);
--red:            #ea3943;   /* Loss, short, negative */
--red-dim:        rgba(234,57,67,0.10);
--amber:          #f0b429;   /* Warning, current price line, approximate values */
--amber-dim:      rgba(240,180,41,0.10);
--blue:           #4e9af1;   /* Informational */

/* Privacy badges */
--badge-zk:      teal        /* ZK-proven exact value */
--badge-dp:      amber       /* Differentially private approximate value */
--badge-public:  text-muted  /* Plaintext on-chain */
--badge-encrypt: teal        /* FHE encrypted field */
```

4.3 Privacy Visual Grammar

Every interface element that touches privacy must follow a consistent visual language. This grammar should be learnable in the first session and automatic by the third.

The FHE Hex Icon (⬡):
A hexagon outline in `--teal` is the universal indicator that a value is FHE-encrypted. It appears next to blurred values, in the navbar FHE status pill, in order confirmation modals, and in the position table. It is never used decoratively.

Blurred values:
Fields that are encrypted but not yet decrypted are shown as blurred text: `filter: blur(4px)`. The underlying text is always 8 characters of random-looking characters rendered at the correct width for the expected value — so "•••••••" blurred looks approximately the same size as "$48,200". This prevents layout shift when decryption occurs.

Decrypted values:
When a value is decrypted, the blur transitions off over 300ms with a fade-in. A brief teal flash (100ms outline glow on the field) confirms the decryption was successful.

Analytics badges:

```
[ZK]     teal pill   — exact value, cryptographically proven
[DP]     amber pill  — approximate value, differentially private
PUBLIC   grey text   — plaintext on-chain, always exact
```

These badges appear inline next to the value label, not after the value. Placement before the value is intentional — it sets expectations before the user reads the number.

4.4 Component Library

Build on a minimal internal component library. Do not use a full UI kit like Radix or Chakra — the aesthetic requirements are specific enough that adapting a generic library creates more friction than building targeted components.

Core components required:

Component	Description
`<PriceDisplay>`	Monospaced price with green/red delta. Animates on update.
`<EncryptedField>`	Blurred value + lock icon. Accepts decrypted value prop to transition.
`<DecryptButton>`	Triggers permit flow. Shows progress states.
`<PrivacyBadge>`	ZK / DP / PUBLIC badge.
`<PoolBadge>`	Pool 1 / Pool 2 indicator.
`<LeverageSelector>`	Preset buttons (1x 2x 5x 10x) + custom input.
`<CollateralInput>`	Amount input with token symbol and max button.
`<OrderSummary>`	Pre-trade summary card with all computed fields.
`<HeatmapBar>`	Single row of the liquidation heatmap.
`<FundingBars>`	Funding history mini chart.
`<VaultUtilMeter>`	Horizontal progress bar for vault utilisation.
`<PermitManager>`	List of active permits with revoke controls.
`<ToastNotification>`	4 severity levels. Auto-dismiss.

4.5 Typography Scale

```
Display:     32px / Geist 500  — page titles, cover numbers
Heading 1:   24px / Geist 500  — section titles
Heading 2:   18px / Geist 500  — panel titles
Heading 3:   14px / Geist 500  — sub-section labels
Body:        14px / Geist 400  — paragraphs, descriptions
Caption:     12px / Geist 400  — labels, table headers, badges
Mono large:  16px / Geist Mono 500  — primary prices, PnL
Mono:        13px / Geist Mono 400  — secondary prices, addresses
Mono small:  11px / Geist Mono 400  — timestamps, minor values
```

4.6 Spacing System

Base unit: 4px. All spacing values are multiples of 4.

```
xs:  4px   — icon gaps, tight inline spacing
sm:  8px   — between related elements
md:  12px  — standard padding inside components
lg:  16px  — standard gap between components
xl:  24px  — section spacing within a panel
2xl: 32px  — panel-to-panel spacing
3xl: 48px  — page-level section separation
```

---

5. Wallet & Authentication Layer

5.1 Supported Wallets

Priority support:

1. MetaMask (injected)
2. Coinbase Wallet
3. WalletConnect v2 (covers 300+ wallets)
4. Rabby (growing in DeFi power-user segment)

Future: Privy or Dynamic for institutional onboarding with email/social fallback.

5.2 Connection Flow

```
Landing state:
  Navbar shows "Connect Wallet" button (ghost style)
  All trading UI is visible but non-interactive (greyed CTA)
  Analytics page is fully accessible without wallet

On connect click:
  Modal opens with wallet options
  No KYC, no email, no geoblock checks at connection time
  (Geoblocking if legally required: handled at API layer, not frontend)

Post-connection:
  Navbar: wallet address shown as 0x4a2f...c91d
  Network check: if wrong chain, show "Switch to Arbitrum" banner
  Pool 2 only: check if operator is set for FHERouter
    → if not set: show "Pool 2 Setup Required" badge until resolved
```

5.3 Pool 2 Operator Setup

Pool 2 requires one-time operator approval before trading or providing liquidity. This is the FHE equivalent of `token.approve()` but for confidential transfers.

UX flow:

```
First time Pool 2 user:

1. User switches to Pool 2 tab
2. Banner appears: "One-time setup required for encrypted trading"
   [Sub-copy]: "Pool 2 uses encrypted token transfers. You need to grant
   ShadeSpot operator permission on your FHE Token once. This replaces
   the standard approve step."
3. Single "Set Up Pool 2" button
4. Wallet signs one transaction: fheToken.setOperator(fheRouter, maxUint48)
5. On success: banner dismisses, Pool 2 fully usable
6. In Settings → Permits: user can see and revoke this permission
```

The operator setup should never appear again after completion (store completion in localStorage keyed by wallet address).

5.4 Chain Management

Target chains at launch:

- Arbitrum Sepolia (testnet, primary)
- Base Sepolia (testnet, secondary)

Production targets (post-buildathon):

- Arbitrum One
- Base

Wrong network banner:

```
[!] You're on Ethereum Mainnet. ShadeSpot runs on Arbitrum.
[Switch to Arbitrum]  [Dismiss]
```

---

6. Pool Selection & Onboarding

6.1 Pool Selector

The pool selector is a persistent top-level choice, not a modal. It appears in the Context Bar on the Trade and Earn pages.

```
┌──────────────────────────┬──────────────────────────────┐
│  ● Pool 1                │    Pool 2                    │
│  USDC collateral         │    FHE Token collateral      │
│  Standard ERC-20         │    Encrypted LP accounting   │
│  LP balance visible      │    LP balance encrypted ⬡    │
└──────────────────────────┴──────────────────────────────┘
```

Pool 1 is selected by default. Pool 2 selection triggers the operator setup check.

6.2 First-Time User Onboarding Modal

On first visit (detected via localStorage), show a one-time modal:

Page 1 of 3 — What is ShadeSpot?

```
ShadeSpot is a perpetual futures exchange where your positions are
encrypted using Fully Homomorphic Encryption.

Nobody can see your position size, entry price, or direction —
not other traders, not bots, not us.

You can decrypt your own data at any time using a cryptographic permit.
```

[Next →]

Page 2 of 3 — Two Pools

```
Pool 1 uses standard USDC as collateral. Your LP position is visible.
This is the familiar DeFi experience, with encrypted trading positions.

Pool 2 uses an FHE Token as collateral. Everything — trading positions
AND LP balances — is encrypted. Maximum privacy.

Both pools share the same markets, funding rates, and liquidation logic.
```

[Next →]

Page 3 of 3 — Reading the Interface

```
  [DP] means the number is approximate (differentially private)
  [ZK] means the number is exact and cryptographically proven
  ⬡    means the field is FHE-encrypted
  •••• means you haven't decrypted this yet — it's yours when you want it

[Start Trading]
```

This modal has a "Don't show again" checkbox and a skip link.

---

7. Trade Page — Pool 1 (USDC)

7.1 Layout

```
┌──────────────────────────────────────────────────────────────────┐
| TICKER BAR                                                        |
| BTC-PERP ▾  |  $83,412  |  +2.34%  |  OI ~$451M [DP]  |  ...  |
├──────────────────────────────────────────┬──────────────────────┤
|                                          |  ORDER PANEL         |
|   CHART                                  |  Market | Limit | Stop |
|   (1H selected)                          |                      |
|   Candlesticks + volume                  |  LONG | SHORT        |
|   Current price dashed line              |  Collateral input    |
|   Liquidation heatmap overlay            |  Leverage selector   |
|   (right edge, bucket-level)             |  Order summary       |
|                                          |  FHE privacy note    |
|                                          |  CTA button          |
|                                          |──────────────────────|
|                                          |  MY POSITION         |
|                                          |  Encrypted fields    |
|                                          |  Decrypt button      |
|                                          |  Decrypted view      |
├──────────────────────────────────────────┴──────────────────────┤
| BOTTOM TABS: Open Positions | Orders | Trade History            |
├──────────────────────────────────────────────────────────────────┤
| ANALYTICS STRIP: Market Stats | Liq. Heatmap | Funding History  |
└──────────────────────────────────────────────────────────────────┘
```

7.2 Ticker Bar

The ticker bar is a horizontal strip showing market-level data. All values carry privacy badges.

Field	Value example	Badge	Source
Mark Price	$83,412.00	PUBLIC	Oracle
24h Change	+$1,904 (+2.34%)	PUBLIC	Oracle delta
Open Interest	~$451M	DP	Encrypted OI + noise
Funding / 8h	+0.0312%	ZK	Proven from encrypted accumulator
24h Volume	$2.31B	ZK	Proven from settlement events
Vault TVL	$28.4M	PUBLIC	Plaintext vault state (Pool 1)
Positions	1,847	PUBLIC	Count only, no details

The open interest badge deserves a tooltip: "This value is differentially private. The true OI is within ±3% of this figure. Individual positions are not revealed." Show on hover.

7.3 Chart

Use `lightweight-charts` (TradingView's open-source library). It handles candlesticks, volume bars, and custom overlays with high performance.

Chart features:

- Candlestick chart with OHLC from price oracle
- Volume bars at the bottom (green/red based on close vs open)
- Current mark price: amber dashed horizontal line with price label
- Index price: teal dotted line (slightly below/above mark)
- Timeframe tabs: 5m, 15m, 1H, 4H, 1D

Liquidation heatmap overlay: A vertical strip on the right edge of the chart (approximately 24px wide) shows bucket-level liquidation concentration. Each bucket is a coloured bar:

- Teal (low count) → Amber (medium) → Red (high concentration)
- Tooltip on hover: "Bucket $73k–$74k: 38 positions"
- Small label "LIQ." at the bottom of the strip
- This uses the public bucket data from the on-chain index — no privacy violation

No individual position markers on chart. Unlike some perp DEXes, we do not show user's own entry price as a line on the chart unless they have explicitly decrypted. If decrypted, show a teal dashed line at entry price with label "My entry $81,340".

7.4 Order Panel

Side toggle: LONG (green) / SHORT (red). Active side gets filled background. Toggle is large enough to tap on mobile.

Order types:

- Market: immediate execution at oracle mark price
- Limit: stored in OrderManager, keeper-executed when price condition met
- Stop: trigger order (short order executing when price drops to trigger)

Collateral input:

```
COLLATERAL
┌──────────────────────────────┐
│  1,000                USDC   │
└──────────────────────────────┘
Balance: 12,450 USDC  [MAX]
```

Leverage selector:

```
[1×]  [2×]  [5×]  [10×]  [Custom]
```

Custom opens an inline input field. Cap enforced at 10x client-side (contract enforces too).

Order summary card:

```
Size              $10,000.00
Entry (est.)      $83,412
Liq. price        $75,071   (shown in red)
Fees (8h)         -$3.12
Liq. bucket       $74k - $75k   (shown in muted)
```

The liquidation bucket field is a deliberate educational moment — it shows the user which public bucket their position will be placed in. Tooltip: "Your liquidation price range is placed in a public bucket so liquidation bots can find your position if needed, without revealing your exact entry or size."

FHE Privacy note (always visible, never dismissible):

```
⬡  Size, collateral & entry encrypted before hitting chain.
   Never visible to other participants.
```

Small teal callout box. This is not an opt-in — it is always on for Pool 1.

CTA button:

- Long: `OPEN LONG · $10,000` (green)
- Short: `OPEN SHORT · $10,000` (red)
- Loading state: spinner + "Encrypting & submitting…"
- Success: button briefly shows checkmark, then resets

Limit order specific: Add a trigger price field between collateral and leverage. Show whether the current oracle price would immediately execute the order (with a warning if so — user probably wants a market order).

7.5 My Position (Right Panel, Below Order Form)

This section shows the user's current position in the selected market. It has two states: encrypted and decrypted.

Encrypted state (default):

```
MY POSITION · BTC-PERP     ⬡ Pool 1

Size          ▬▬▬▬▬▬▬▬  ⬡
Collateral    ▬▬▬▬▬▬▬▬  ⬡
Entry         ▬▬▬▬▬▬▬▬  ⬡

[ DECRYPT VIA PERMIT · ~1.4s ]
Threshold Network decryption
```

The blurred values are shown with correct-width blur masks. The user should be able to see that values exist without seeing what they are.

Loading state (after decrypt click):

```
MY POSITION · BTC-PERP

⬡  Requesting decryption from Threshold Network…
   [▬▬▬▬▬▬▬▬
```

```
MY POSITION · BTC-PERP     ✓ Decrypted

DIRECTION     LONG ↑
SIZE          $48,200
COLLATERAL    $4,820
ENTRY         $81,340
UNREAL. PNL   +$1,024.40   ↑
LIQ. PRICE    $73,206
FUNDING PAID  -$18.44

[CLOSE POSITION]  [SHARE PERMIT ↗]
```

The "SHARE PERMIT" link opens the selective disclosure flow (see section 9).

Permit expiry indicator: If a permit exists but is approaching expiry, show:

```
▲ Permit expires in 2h 14m   [Renew]
```

---

8. Trade Page — Pool 2 (FHE Token)

8.1 Differences from Pool 1

Pool 2 uses the same layout as Pool 1 with these differences:

Collateral input shows FHE Token:

```
COLLATERAL
┌──────────────────────────────┐
│  1,000            FHE Token ⬡  │
└──────────────────────────────┘
Balance: ▬▬▬▬▬▬ ⬡   [DECRYPT TO SEE]
```

The user's FHE Token balance is encrypted. The balance display shows a blurred value with a small "decrypt to see" link. Clicking it decrypts just the balance (separate lightweight permit, or reuses existing).

Vault TVL in ticker bar: Pool 2 vault TVL is encrypted. Show as:

```
Vault TVL   ▬▬▬ ⬡
```

With tooltip: "Pool 2 vault depth is encrypted. Individual LP balances are not visible to anyone on-chain."

Operator status indicator: In the order panel header, show a small indicator:

```
⬡ Operator: Active  [Manage]
```

If not active, show the setup prompt inline.

FHE Privacy note is stronger:

```
⬡  Pool 2: Both your trading position AND the vault's LP accounting
   are encrypted. This is the maximum privacy configuration.
```

8.2 FHE Token Balance Decryption

When the user clicks "DECRYPT TO SEE" next to their balance:

```
1. Check if a valid balance permit exists (cached in memory / localStorage)
2. If yes: decrypt inline, show value with teal flash confirmation
3. If no: run abbreviated permit flow (just for balance field)
   — lighter UX than full position decrypt
   — single "Decrypt Balance" button, ~0.8s
```

Balance decryption is lighter than position decryption because it only requests a single euint64 field rather than a full position struct.

---

9. Position Management & Decryption UX

9.1 The /positions Page

A dedicated page showing all open positions across both pools. This is the primary position management view for power users.

```
MY POSITIONS                                2 active · all encrypted

Pool selector: [All] [Pool 1] [Pool 2]

┌──────┬──────┬─────────┬──────────┬────────┬──────┬──────┬──────────┬──────────┐
|MARKET| POOL |  SIDE   |   SIZE   |COLLAT. |ENTRY | MARK |    PNL
|LIQ.BUCKET|          |
├──────┼──────┼─────────┼──────────┼────────┼──────┼──────┼──────────┼──────────┤
|BTC   |  1   | LONG ↑  | ▬▬▬ ⬡   | ▬▬▬ ⬡| ▬▬▬ |83,412|+$1,024.40 |$73k-$74k
|[DECRYPT] |
|ETH   |  1   | SHORT ↓ | ▬▬▬ ⬡   | ▬▬▬ ⬡| ▬▬▬ | 1,884| +$281.00  |$2.1k-
$2.2k|[DECRYPT]|
└──────┴──────┴─────────┴──────────┴────────┴──────┴──────┴──────────┴──────────┘
```

Key design notes:

- MARK price is always shown (public oracle data)
- PnL is shown in plaintext — it is derived from mark price vs encrypted entry price, computed by the frontend using oracle price only (approximate). Exact PnL requires decryption.
- "Approximate PnL" badge appears next to the value until decrypted: `~+$1,024`
- After decryption, the `~` is removed and the exact value is shown
- LIQ. BUCKET is always shown (public on-chain data)
- SIDE is always shown because it is not stored as a secret — the position key encodes direction

Wait — SIDE being public deserves a design decision note: the position key is `keccak256(trader, indexToken, isLong)`. If the position exists, the existence and direction are inferable from the key. We should not pretend direction is private when it is technically inferable. Show it honestly. This aligns with the "label the privacy guarantee" mandate.

9.2 Decryption Flow — Full Position

```
User clicks [DECRYPT] on a position row:

Step 1 — Check permit cache
  If valid permit exists in memory: skip to Step 3
  If permit in localStorage but near expiry: show renewal prompt
  If no permit: Step 2

Step 2 — Permit creation
  Modal appears:
  ┌──────────────────────────────────────────┐
  |  Decrypt Position                         |
  |                                           |
  |  Sign a permit to view your BTC-PERP LONG  |
  |  position. This does not reveal your data  |
  |  to anyone else.                          |
  |                                           |
  |  Access expires: [1 hour ▾]               |
  |  (Options: 15 min, 1 hour, 8 hours, custom) |
  |                                           |
  |  [Sign Permit]  [Cancel]                  |
  └──────────────────────────────────────────┘

  Wallet signs EIP-712 typed data (no gas, off-chain signature)

Step 3 — Threshold Network request
  Status shown inline on the position row:
  "⬡ Requesting from Threshold Network… (1/3 signatures)"
```

```
  "⬡ Requesting from Threshold Network… (2/3 signatures)"
  "⬡ Requesting from Threshold Network… (3/3 signatures)"

  ~1.2–1.8 seconds total

Step 4 — Reveal
  All encrypted fields in the row animate from blurred to clear (300ms)
  Teal flash on each field as it reveals
  Row expands slightly to show exact values

  Size:       $48,200.00
  Collateral: $4,820.00
  Entry:      $81,340.00
  PnL:        +$1,024.40  (exact, not approximate)
```

9.3 Selective Disclosure

The selective disclosure flow allows a user to grant another address the ability to view their position. This is the institutional compliance feature.

Access point: From a decrypted position row, click "Share Permit ↗"

```
SHARE POSITION ACCESS

Grant another address permission to view this position.
They will see the same decrypted data you see.

Recipient address
┌──────────────────────────────────────────┐
│  0x...                                   │
└──────────────────────────────────────────┘

Access level:   [View only ▾]
Expiry:         [24 hours ▾]

⚠ The recipient will be able to see: size, collateral,
  entry price. They will NOT be able to trade or modify
  your position.

[Grant Access]  [Cancel]
```

Active permits panel (in Settings → Permits):

```
ACTIVE PERMITS

Position: BTC-PERP LONG (Pool 1)
  Granted to: 0x9f2a...4b1c   (labelled "My Auditor")
  Expires: 23h 41m remaining
  [Revoke]

Position: ETH-PERP SHORT (Pool 1)
  Granted to: 0x7e3b...8d2f   (unlabelled)
  Expires: 6h 12m remaining
  [Revoke]

Balance: FHE Token (Pool 2)
  Granted to: 0x4a2f...c91d   (self — via decrypt)
  Expires: 47m remaining
  [Renew]  [Revoke]
```

Recipients can label their grants. Labels stored in localStorage (off-chain, private).

---

10. Order Management

10.1 Orders Tab

Below the chart, the Orders tab shows pending limit and stop orders.

```
ORDERS                                                3 pending


┌────────┬────────┬──────────┬────────────────┬──────────────┬────────┬──────────┐
│MARKET  │ TYPE   │  SIDE    │  COLLATERAL    │  TRIGGER     │LEVERAGE│          │
├────────┼────────┼──────────┼────────────────┼──────────────┼────────┼──────────┤
│BTC     │ LIMIT  │ LONG ↑   │ ▬▬▬ ⬡          │ ▬▬▬ ⬡        │ 10x    │[CANCEL]  │
│ETH     │ STOP   │ SHORT ↓  │ ▬▬▬ ⬡          │ ▬▬▬ ⬡        │ 5x     │[CANCEL]  │
│BTC     │ LIMIT  │ SHORT ↓  │ ▬▬▬ ⬡          │ ▬▬▬ ⬡        │ 10x    │[CANCEL]  │
└────────┴────────┴──────────┴────────────────┴──────────────┴────────┴──────────┘
```

Both collateral amount and trigger price are stored as `euint128` on-chain (OrderManager). The frontend shows them as blurred by default with a per-row decrypt option.

Trigger price is particularly sensitive — showing it would tell observers exactly where you expect the market to move.

Cancel order flow:

Pool 1 orders: standard transaction via Router.cancelOrder(), collateral refunded immediately.

No confirmation modal required for cancel (it is reversible — they can re-create the order). Just show a brief loading state on the cancel button, then remove the row on success with a toast: "Order cancelled. Collateral returned."

10.2 Order Execution Feed

A subtle real-time feed in the bottom of the Orders tab shows recent keeper executions (all users, anonymised):

```
RECENT EXECUTIONS
  BTC-PERP LONG filled at $83,210    2s ago
  ETH-PERP SHORT filled at $1,878    14s ago
  BTC-PERP LONG filled at $83,190    31s ago
```

No size or wallet shown — just market, direction, and fill price. This gives traders confidence that the keeper network is active.

---

11. Liquidation UX

11.1 Liquidation Risk Indicator

On the right panel, below the position data (if decrypted), show a visual liquidation distance indicator:

```
LIQUIDATION RISK

Current price    $83,412
Your liq. price  $73,206

Distance: 12.2% below current price

[▓▓▓
```

```
⚠ Liquidation Warning
BTC-PERP LONG is approaching liquidation.
Distance: 8.3% ($5,842)
[View Position]  [Add Margin — coming soon]
```

Orange toast, auto-dismisses after 10 seconds, reappears if price continues moving toward liquidation.

If position is encrypted and user has not decrypted, we cannot compute this warning. Show instead:

```
ℹ Decrypt your positions to receive liquidation alerts.
[Decrypt Now]
```

This is a gentle incentive to decrypt — the user gains something tangible (safety alerts) in exchange for the brief ~1.5 second decryption step.

11.3 Getting Liquidated

If a user's position is liquidated (detected via PositionFlagged and LiquidationExecuted events):

```
[Modal, cannot be dismissed until acknowledged]

Your BTC-PERP LONG position was liquidated.

Your position was underwater and a liquidation bot closed it.

Liquidator received 5% of your collateral as a reward.

This position's data has been cleared from the chain.
You can view historical position data in Trade History.

[I Understand]
```

The modal is sobering, not alarming. No dramatic red animations. Just clear information.

---

12. Earn Page — Liquidity Provision

12.1 Layout

```
┌──────────────────────────────────────────────────────────┐
| EARN                                                      |
| Provide liquidity and earn trading fees                   |
├──────────────────────┬───────────────────────────────────┤
```

```
|  POOL SELECTOR     |  VAULT STATS                        |
|  [Pool 1] [Pool 2] |  TVL / Util / 7d APY / 30d APY   |
├────────────────────┴─────────────────────────────────────┤
|  ┌──────────────────────┐  ┌──────────────────────────┐ |
|  |  ADD LIQUIDITY       |  |  MY POSITION             | |
|  |  Amount input        |  |  (blurred or decrypted)  | |
|  |  Pool info           |  |  Decrypt button          | |
|  |  [Add Liquidity]     |  |  [Remove Liquidity]      | |
|  └──────────────────────┘  └──────────────────────────┘ |
├──────────────────────────────────────────────────────────┤
|  POOL PERFORMANCE                                         |
|  Revenue chart / Trader PnL distribution / Fee history |
└──────────────────────────────────────────────────────────┘
```

12.2 Pool 1 LP Experience

Pool 1 LP positions use plaintext `uint256` accounting. The LP's share is visible directly.

Vault stats:

```
Pool 1 Vault (USDC)

TVL             $28,400,000   PUBLIC
Utilisation     67.4%         PUBLIC
Reserved        $19,140,000   PUBLIC
Available       $9,260,000    PUBLIC
7d APY          8.2%          PUBLIC
30d APY         11.4%         PUBLIC
```

My LP position:

```
MY POOL 1 POSITION

Your share       $12,430.00
Pool share %     0.044%
Est. daily yield $2.80
Since deposit    +$430.20 (+3.6%)

[Remove Liquidity]
```

The LP risk disclosure must be visible on this page:

```
ℹ LP Risk
As a liquidity provider, you are the counterparty to all traders.
If traders are net profitable, your pool balance decreases.
If traders are net losing, your pool balance increases.
Vault utilisation above 80% means new positions cannot open
and withdrawals are restricted until utilisation drops.
```

12.3 Pool 2 LP Experience

Pool 2 LP positions are stored as `euint64` ciphertexts. The LP balance is encrypted.

Vault stats:

```
Pool 2 Vault (FHE Token)

TVL             ▬▬▬ ⬡     ENCRYPTED
Utilisation     ▬▬▬ ⬡     ENCRYPTED
7d APY          ~8.4%      [DP] estimated
```

Note: TVL for Pool 2 is encrypted. APY is estimated from fee events (not exact) and carries a DP badge.

My LP position:

```
MY POOL 2 POSITION  ⬡

Your share       ▬▬▬▬▬▬  ⬡
Pool share %     ▬▬▬▬▬▬  ⬡
Est. daily yield ▬▬▬▬▬▬  ⬡

[ DECRYPT MY POSITION · ~1.2s ]
```

After decryption:

```
MY POOL 2 POSITION  ✓ Decrypted

Your share       2,430.00 FHE Token
Pool share %     —  (pool total encrypted, ratio unavailable)
Est. daily yield ~0.56 FHE Token / day  [DP]

[Remove Liquidity]  [Share Permit ↗]
```

Note that pool share percentage cannot be computed without knowing the total pool size, which is encrypted. This is an honest limitation — do not fake the number. Show "—" with a tooltip:

"Pool 2's total liquidity is encrypted. Your exact share percentage is not computable without decrypting the pool total."

---

13. Analytics Page

13.1 Philosophy

The analytics page embodies the "privacy-preserving analytics" narrative. Every metric is clearly labelled with its privacy model. Users can understand both what the market looks like AND why they can trust the numbers despite the underlying data being encrypted.

This page is fully accessible without a wallet connection.

13.2 Layout

```
ANALYTICS

Market: [BTC-PERP ▾]   Pool: [All ▾]   Timeframe: [24H ▾]

┌──────────────────┬──────────────────┬──────────────────┬────────────────┐
│  LONG/SHORT      │  FUNDING RATE    │  24H VOLUME      │  OPEN INT.     │
│  58 / 42  [DP]   │  +0.031% [ZK]    │  $2.31B  [ZK]    │  ~$451M [DP]   │
│  longs domin.    │  longs pay       │  from settl.     │  ±3% noise     │
└──────────────────┴──────────────────┴──────────────────┴────────────────┘

┌──────────────────────────────────────────┬──────────────────────────┐
│  LIQUIDATION HEATMAP                     │  FUNDING HISTORY         │
│  (bucket-level · proven)                 │  (8h epochs · 7 days)    │
│                                          │                          │
│  $86k-87k  ▒▒▒
```

```
┌──────────────────────────────────────────────────────────────────┐
│  HOW THESE NUMBERS ARE COMPUTED                                  │
│  [Expandable explanation section — see 13.4]                     │
└──────────────────────────────────────────────────────────────────┘
```

13.3 Privacy Badge Legend (Persistent on Analytics Page)

At the top of the analytics page, always visible:

```
About these numbers:
  [ZK]  Exact value, verified by zero-knowledge proof. Cryptographically correct.
  [DP]  Approximate value. Individual positions are not revealed.
        Differential privacy adds controlled noise. Accuracy: ±3–5%.
```

13.4 "How These Numbers Are Computed" — Expandable Section

This section is critical for the institutional narrative. It should be clear, non-technical in tone, but specific enough to satisfy a sophisticated reader.

```
HOW THESE NUMBERS ARE COMPUTED

Volume [ZK]
When a position closes, the settlement amount is transferred on-chain as
plaintext.
ShadeSpot aggregates these settlement events and publishes the total. A zero-
knowledge
proof verifies the aggregate is correct. Your individual trade size is never
revealed.

Funding Rate [ZK]
The funding rate is computed from the imbalance between total long and short open
interest.
Both OI figures are encrypted on-chain. The funding rate computation runs inside
the FHE
coprocessor and the resulting rate is verified by a zero-knowledge proof before
publication.
The exact OI figures remain encrypted.

Long/Short Ratio [DP]
The ratio of long to short open interest is published with calibrated statistical
noise
added (differential privacy). The true ratio is within ±5% of the displayed value.
This approach ensures individual position sizes cannot be reverse-engineered from
the aggregate.

Open Interest [DP]
Approximate total open interest. Same differential privacy approach as L/S ratio.

Liquidation Heatmap
Position count per $1,000 price bucket. Bucket assignment is public (required for
liquidation bots to discover underwater positions). No individual position details
are
revealed — only the count of positions in each bucket.
```

---

14. Governance Page

14.1 Scope

At launch, governance is minimal. The page exists to establish the legitimacy of the governance model and to track protocol parameter changes.

What can be governed (post-launch):

- Vault utilisation cap

- Liquidation threshold (currently 80%)
- Liquidator reward (currently 5%)
- Funding rate precision
- Bucket size per market
- Oracle migration (owner-set → Pyth/Chainlink)
- Fee parameters

14.2 Layout

```
GOVERNANCE

Active Proposals (0)
No active proposals.

Recent Decisions
  [Resolved]  Increase max leverage to 20x on ETH-PERP
  Result: Rejected · 78% against
  Closed: 14 days ago

Parameter Registry
  Max leverage          10x
  Liquidation threshold 80%
  Liquidator reward     5%
  Funding interval      1 hour
  Bucket size (BTC)     $1,000
  Oracle                Owner-set (Pyth migration planned)

Propose a Change
  [Connect wallet to propose]
```

Governance interface is intentionally minimal at launch. Expand in Phase 3.

---

15. Privacy Centre

15.1 Purpose

The Privacy Centre is a standalone page that explains ShadeSpot's privacy model to a non-cryptographer. It serves three audiences: retail users who want to understand what "encrypted" means, institutional users who need to understand the guarantees for compliance, and curious developers who want to dig into the architecture.

15.2 Structure

```
PRIVACY CENTRE

Your positions are encrypted using Fully Homomorphic Encryption.
Here is exactly what that means.

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

WHAT IS ENCRYPTED

  ✓ Position size
  ✓ Collateral amount
  ✓ Entry price
  ✓ Pool 2 LP balance
  ✓ Pool 2 total vault depth

WHAT IS NOT ENCRYPTED (and why)

  × That your position exists
      (Position keys are public — required for liquidation)
  × Which direction (long/short)
      (Inferable from position key — we do not pretend otherwise)
  × Which price bucket your liquidation falls in
      (Required for liquidation bot discovery)
  × Your net PnL when you close
      (Token transfers require plaintext amounts)
  × Oracle prices
      (Market prices are public by nature)

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━

HOW YOUR DATA IS KEPT PRIVATE

  FHE (Fully Homomorphic Encryption) allows the protocol to
  compute directly on your encrypted data — checking if your
  position is underwater, calculating your PnL, applying
  funding fees — without ever decrypting it.

  The only party that can decrypt your position is you,
  using a cryptographic permit signed from your wallet.

  [Learn more about FHE →] (links to Fhenix docs)

━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
```

SELECTIVE DISCLOSURE

You can grant another address temporary permission to view your position. This is useful for:

- Auditors verifying your risk exposure
- Counterparties confirming a trade
- Tax reporting services

Permits expire and can be revoked at any time from Settings → Permits.

[Manage Your Permits →]

WHO RUNS THE PRIVACY INFRASTRUCTURE

ShadeSpot uses Fhenix CoFHE for on-chain FHE computation. Decryption is handled by the Fhenix Threshold Network — a distributed set of keyholders where no single party can decrypt your data alone. A threshold (majority) of keyholders must cooperate, and they only respond to signed permits from your wallet.

The ShadeSpot team does not have access to your position data.

THE ANALYTICS TRADE-OFF

To preserve privacy while providing market transparency, ShadeSpot uses two techniques:

Zero-knowledge proofs verify that aggregate statistics (volume, funding rate) are correct without revealing the underlying positions.

Differential privacy adds controlled noise to statistics derived from encrypted data (open interest, L/S ratio) so individual positions cannot be reverse-engineered.

[Read more about the analytics approach →]

SECURITY & AUDITS

```
  Smart contract audit: [Pending]
  FHE library audit: Handled by Fhenix (link)
  Bug bounty: [Coming soon]

  [View Contracts on Arbiscan →]
```

---

16. Notification & Alert System

16.1 Toast Types

```
SUCCESS (green left border):
  "Position opened. BTC-PERP LONG $10,000"
  "Permit granted to 0x9f2a...4b1c"
  "Liquidity added: $5,000 to Pool 1"

INFO (blue left border):
  "Funding rate updated: +0.031% / 8h"
  "New position flagged in your liquidation range"

WARNING (amber left border):
  "BTC-PERP LONG approaching liquidation (8.3% away)"
  "Permit expires in 30 minutes"
  "Vault utilisation above 90% — new positions may be restricted"

ERROR (red left border):
  "Transaction failed: Insufficient collateral"
  "Decryption failed. Threshold Network timeout. Try again."
  "Oracle price stale — trading paused"
```

Toasts stack in the top-right corner, newest on top. Auto-dismiss after 6 seconds for success/info, 10 seconds for warning, manual dismiss only for errors.

16.2 Persistent Alert Banner

For protocol-wide issues (oracle staleness, coprocessor unavailability):

```
[!] The FHE coprocessor is experiencing delays. Liquidation checks
    may be slow. Trading is unaffected.
```

Full-width banner below navbar, amber background, dismissible after reading. Re-appears if condition persists for >5 minutes.

16.3 Liquidation Proximity Alert

Real-time calculation (only for decrypted positions): when mark price moves within 15% of a liquidation price, send an in-app notification and an optional browser notification (with permission).

Browser notification (if permitted):

```
ShadeSpot — Liquidation Warning
Your BTC-PERP LONG is 8.3% from liquidation ($73,206).
Current price: $79,503.
```

---

17. State Management Architecture

17.1 Technology Choice

Use **Zustand** for global state management. It is minimal, TypeScript-friendly, and does not require provider wrapping. Avoid Redux — the complexity is not warranted for this application.

Use **React Query (TanStack Query)** for server/chain state — oracle prices, vault stats, position handles, analytics data. It handles caching, background refresh, and loading states cleanly.

17.2 Zustand Stores

```typescript
```

```typescript
// walletStore.ts
interface WalletStore {
  address: string | null
  chainId: number | null
  isConnected: boolean
  activePool: 1 | 2
  isPool2OperatorSet: boolean
  connect: () => Promise<void>
  disconnect: () => void
  setPool: (pool: 1 | 2) => void
}

// positionStore.ts
interface PositionStore {
  positions: Record<string, EncryptedPosition>  // key → ciphertext handles
  decryptedPositions: Record<string, DecryptedPosition>  // key → plaintext
  permits: Record<string, PermitData>  // key → permit + expiry
  decryptPosition: (key: string) => Promise<void>
  grantPermit: (key: string, recipient: string, expiry: number) => Promise<void>
  revokePermit: (key: string, recipient: string) => void
}

// marketStore.ts
interface MarketStore {
  selectedMarket: string  // 'BTC-PERP'
  prices: Record<string, number>  // token → oracle price
  lastPriceUpdate: Record<string, number>  // token → timestamp
  subscribeToPrice: (token: string) => () => void
}

// notificationStore.ts
interface NotificationStore {
  toasts: Toast[]
  alerts: Alert[]
  addToast: (toast: Omit<Toast, 'id'>) => void
  dismissToast: (id: string) => void
  addAlert: (alert: Alert) => void
  dismissAlert: (id: string) => void
}
```

17.3 React Query Keys

```typescript
```

```
const queryKeys = {
  vaultStats: (pool: 1 | 2) => ['vault', pool, 'stats'],
  oraclePrice: (token: string) => ['oracle', token, 'price'],
  openPositions: (address: string) => ['positions', address, 'open'],
  orderBook: (address: string) => ['orders', address],
  analyticsOI: (market: string) => ['analytics', market, 'oi'],
  analyticsFunding: (market: string) => ['analytics', market, 'funding'],
  heatmap: (market: string) => ['analytics', market, 'heatmap'],
  lpBalance: (address: string, pool: 1 | 2) => ['lp', address, pool, 'balance'],
}
```

17.4 Price Subscription

Oracle prices must update in real-time. Use WebSocket subscription to an on-chain event listener:

```typescript
// Subscribe to PriceUpdated events
const unsubscribe = subscribeToOraclePrice(token, (price) => {
  marketStore.getState().updatePrice(token, price)

  // Check liquidation proximity for all decrypted positions
  checkLiquidationProximity(price, token)
})
```

Price updates also trigger background liquidation checks for all decrypted positions — recompute PnL and liquidation distance on every oracle tick.

---

18. CoFHE SDK Integration

18.1 SDK Initialisation

```typescript
```

```
// cofhe.ts — initialise once at app load
import { createCofheConfig, createCofheClient } from '@cofhe/sdk/web'
import { chains } from '@cofhe/sdk/chains'

const config = createCofheConfig({
  supportedChains: [chains.arbitrumSepolia, chains.baseSepolia],
})

export const cofheClient = createCofheClient(config)

// In App.tsx or a provider:
await cofheClient.connect(publicClient, walletClient)
```

18.2 Encrypting Position Inputs

Before calling `Router.openPosition()`, all sensitive fields must be encrypted client-side:

```typescript
import { Encryptable, FheTypes } from '@cofhe/sdk'

async function encryptPositionInputs(
  size: bigint,
  collateral: bigint,
  entryPrice: bigint,
  isLong: boolean
) {
  const [encSize, encCollateral, encEntryPrice, encIsLong] = await cofheClient
    .encryptInputs([
      Encryptable.uint128(size),
      Encryptable.uint128(collateral),
      Encryptable.uint128(entryPrice),
      Encryptable.bool(isLong),
    ])
    .execute()

  return { encSize, encCollateral, encEntryPrice, encIsLong }
}
```

18.3 Decrypting a Position

```typescript
```

```typescript
async function decryptPosition(positionKey: string, permit: PermitData) {
  const handles = await getPositionHandles(positionKey)  // from contract

  const [size, collateral, entryPrice] = await Promise.all([
    cofheClient.decryptForView(handles.size, FheTypes.Uint128).execute(),
    cofheClient.decryptForView(handles.collateral, FheTypes.Uint128).execute(),
    cofheClient.decryptForView(handles.entryPrice, FheTypes.Uint128).execute(),
  ])

  return {
    size: formatUnits(size, 6),       // USDC 6 decimals
    collateral: formatUnits(collateral, 6),
    entryPrice: formatUnits(entryPrice, 18),  // price in 1e18
  }
}
```

18.4 Pool 2 Balance Decryption

```typescript
async function decryptFHEBalance(address: string) {
  const handle = await fheVault.getEncryptedBalance(address)

  await cofheClient.permits.getOrCreateSelfPermit()

  const balance = await cofheClient
    .decryptForView(handle, FheTypes.Uint64)
    .execute()

  return formatUnits(balance, 18)  // FHE Token decimals
}
```

18.5 Permit Creation for Selective Disclosure

```typescript
```

```
async function createSelectiveDisclosurePermit(
  positionKey: string,
  recipientAddress: string,
  expiryTimestamp: number
) {
  const permit = await cofheClient.permits.createPermit({
    contracts: [POSITION_MANAGER_ADDRESS],
    recipient: recipientAddress,
    expiry: expiryTimestamp,
  })

  // Store permit reference for management UI
  permitStore.addPermit(positionKey, recipientAddress, permit)

  return permit
}
```

---

19. Performance & Loading States

19.1 Loading Hierarchy

The application has distinct loading states at different levels:

Level 1 — App shell: Navbar, sidebar, and layout skeleton render immediately. Never block on wallet connection or data fetching.

Level 2 — Market data: Chart and ticker bar show skeleton loaders while prices load (~100ms on warm cache). Use lightweight-charts' built-in loading state.

Level 3 — Position data: Position table shows rows with blurred values immediately (the blurred state IS the loaded state). No additional loading indicator needed — blurred = encrypted, not loading.

Level 4 — Decryption: Only the decrypt action has a loading state. Show progress with step indicators.

19.2 Optimistic Updates

For order placement and cancellation, use optimistic updates:

```
User clicks "Open Long":
  Immediately add a pending row to the positions table
  Row shows "Pending…" in the status column
  Actual transaction confirms within 1-3 seconds
  On confirmation: row updates to encrypted state (blurred)
  On failure: row removed, error toast shown
```

19.3 Price Animation

When oracle prices update, animate the change:

- Price increases: briefly flash green background on the price cell
- Price decreases: briefly flash red background
- Duration: 400ms
- Use CSS transitions, not JS timers

Mark price in the ticker bar should update smoothly without layout shift. Monospaced font ensures consistent character widths.

---

20. Error Handling & Edge Cases

20.1 Error Categories

User errors (recoverable):

- Insufficient collateral: show inline error below collateral input
- Leverage too high: inline error, suggest lower leverage
- Wrong order type for market conditions: warning with suggestion

Network errors (retryable):

- RPC timeout: "Transaction may have been submitted. Check transaction history." with retry button
- Oracle stale: banner alert, disable order submission, show staleness duration

Decryption errors:

- Threshold Network timeout: "Decryption timed out. The Threshold Network may be congested. Try again in a moment." with retry button
- Invalid permit: "Your permit has expired or been revoked. Sign a new permit to decrypt." with new-permit button
- Wrong network for decryption: "Connect to Arbitrum to decrypt your positions."

Pool 2 specific:

- Operator not set: inline prompt in order panel, link to setup
- Confidential transfer insufficient balance: "Insufficient FHE Token balance. Your balance is encrypted — decrypt to confirm."

- Encrypted solvency check failed: "Trade would exceed vault capacity." (vague by necessity — Pool 2 vault depth is encrypted)

20.2 Vault at 100% Utilisation

When `availableLiquidity === 0`:

```
▲ Pool 1 is at 100% utilisation.
New positions cannot be opened and withdrawals are restricted
until traders close positions or additional liquidity is added.

[Provide Liquidity →]
```

Disable the order form CTA completely (do not just grey it — remove it and replace with the above message).

20.3 Coprocessor Unavailability

If the FHE coprocessor (responsible for liquidation scanning) has not posted flags in over 30 minutes:

```
[!] Liquidation scanning is delayed (27 minutes since last scan).
    The challenge mechanism will activate after 50 blocks.
    Trading is unaffected. Monitor the status page for updates.

[Status Page →]
```

This is important: traders should know that the safety net for liquidations is temporarily degraded, even if their trade execution is unaffected.

---

21. Responsive & Mobile

21.1 Breakpoints

```
Mobile:  < 768px
Tablet:  768px – 1199px
Desktop: ≥ 1200px
Large:   ≥ 1600px  (full analytics strip visible)
```

21.2 Mobile Trade Page

On mobile, the trade page collapses to a tab-based layout:

```
[Chart] [Trade] [Positions] [Analytics]
```

Each tab shows its section full-screen. The chart tab shows price only (no embedded order form). The Trade tab shows the full order form. The Positions tab shows the positions table.

Key mobile adaptations:

- Long/Short toggle is full-width, easy to tap
- Leverage selector uses larger tap targets
- Decrypt button is full-width in position rows
- Toasts appear at the bottom of screen on mobile (not top-right)

21.3 Tablet Layout

On tablet, show chart + order panel side by side (same as desktop) but hide the analytics strip. The analytics strip is accessible via a slide-up drawer triggered by an "Analytics" button at the bottom of the page.

---

22. Accessibility

22.1 Core Requirements

- All interactive elements must be keyboard-accessible
- Color is never the only indicator (use shape/text alongside color)
- Screen reader compatibility for critical actions (position open, close, decrypt)
- ARIA labels on all icon-only buttons
- Focus management in modals (trap focus, return focus on close)
- Sufficient colour contrast (WCAG AA minimum: 4.5:1 for body text)

22.2 Colour Contrast Notes

The dark theme at `#07090d` background with `#7a8fa8` text-secondary gives a contrast ratio of approximately 4.6:1 — just above the WCAG AA threshold. Do not darken `text-secondary` further for aesthetic reasons.

`--teal (#06d6a0)` on `--bg-surface (#0c0f15)` gives a contrast ratio of approximately 9.2:1 — excellent.

22.3 Blurred Values & Screen Readers

Blurred values pose a specific accessibility challenge. Screen readers should read "Encrypted value" rather than attempting to read the blurred text. Implement:

```html
```

```
<span class="encrypted-field" aria-label="Encrypted value">
  <span aria-hidden="true" class="blur">$48,200</span>
</span>
```

After decryption, swap the aria-label to the actual value:

```html
<span aria-label="$48,200">$48,200</span>
```

---

23. Security Considerations

23.1 Permit Storage

Permits are sensitive data. If a permit is stolen, the holder can read the position data until the permit expires.

- Store active permits in memory only (React state, Zustand store). Do not persist to localStorage.
- On tab close: permits are lost. Users must re-decrypt on next session. This is a deliberate privacy trade-off.
- On wallet disconnect: clear all permits immediately.
- On permit grant to third party: log the grant in localStorage (to enable management UI) but do not store the permit data itself.

23.2 Input Validation

All numeric inputs must be validated client-side before encryption:

- Collateral: must be > 0, must be ≤ user balance, must be in valid decimal range for token
- Leverage: must be 1–10 integer
- Trigger price: must be > 0, must be realistic relative to current oracle price (reject trigger prices 10x+ away from market — likely user error)

Do not rely solely on contract validation. Front-end validation provides faster feedback and prevents unnecessary gas costs from failed transactions.

23.3 Oracle Price Sanity Checks

Before displaying or using any oracle price, validate:

- Price > 0
- Price is within 5% of last known price (reject anomalous prices)

- Price was updated within `MAX_PRICE_DELAY` (5 minutes)

If price fails validation, show staleness banner and disable order submission. Never allow orders to be placed against stale prices.

---

24. Environment & Deployment

24.1 Environment Variables

```bash
# Chain
NEXT_PUBLIC_CHAIN_ID=421614           # Arbitrum Sepolia
NEXT_PUBLIC_RPC_URL=https://...       # Alchemy or Infura endpoint
NEXT_PUBLIC_WS_URL=wss://...          # WebSocket for price subscriptions

# Contracts (Pool 1)
NEXT_PUBLIC_ROUTER_ADDRESS=0x...
NEXT_PUBLIC_VAULT_ADDRESS=0x...
NEXT_PUBLIC_POSITION_MANAGER=0x...
NEXT_PUBLIC_FUNDING_MANAGER=0x...
NEXT_PUBLIC_LIQUIDATION_MANAGER=0x...
NEXT_PUBLIC_ORACLE_ADDRESS=0x...
NEXT_PUBLIC_USDC_ADDRESS=0x...

# Contracts (Pool 2)
NEXT_PUBLIC_FHE_ROUTER_ADDRESS=0x...
NEXT_PUBLIC_FHE_VAULT_ADDRESS=0x...
NEXT_PUBLIC_FHE_TOKEN_ADDRESS=0x...

# CoFHE
NEXT_PUBLIC_COFHE_CHAIN=arbitrum-sepolia

# Analytics
NEXT_PUBLIC_ANALYTICS_INDEXER=https://...  # Off-chain indexer API
```

24.2 Tech Stack

```
Framework:     Next.js 14 (App Router)
Language:      TypeScript (strict mode)
State:         Zustand + TanStack Query
Chain:         viem + wagmi v2
FHE:           @cofhe/sdk, @cofhe/react
Charts:        lightweight-charts (TradingView)
Styling:       Tailwind CSS (for layout/utility) + CSS variables (for theming)
```

```
Fonts:          Geist Mono + Geist (via next/font)
Testing:        Vitest + Testing Library
Linting:        ESLint + Prettier
Deployment:     Vercel
```

24.3 Key Third-Party Dependencies

Package	Purpose	Version
`@cofhe/sdk`	FHE encryption/decryption	latest
`@cofhe/react`	React hooks (useEncrypt, useDecrypt)	latest
`wagmi`	Wallet connection	v2
`viem`	Ethereum interactions	v2
`lightweight-charts`	Price chart	^4
`zustand`	Global state	^4
`@tanstack/react-query`	Server state	^5
`next`	Framework	14

Appendix A — Page-by-Page Summary

Page	Auth required	FHE interaction	Key privacy elements
/trade (Pool 1)	No (view) / Yes (trade)	Encrypt inputs, decrypt own position	Blurred positions, FHE note, bucket field
/trade (Pool 2)	Yes (operator set required)	Encrypt inputs, decrypt balance + position	Encrypted balance, FHE vault stats
/positions	Yes	Decrypt all positions	Batch decrypt, permit management
/history	Yes (optional decrypt)	Decrypt historical positions	Closed positions are plaintext PnL; open details encrypted
/earn (Pool 1)	No (view) / Yes (LP)	None	Plaintext LP position
/earn (Pool 2)	Yes	Decrypt LP balance	Blurred TVL, encrypted LP share

Page	Auth required	FHE interaction	Key privacy elements
/analytics	No	None	DP/ZK badges, heatmap, methodology
/govern	No (view) / Yes (propose)	None	None
/privacy	No	None	Explanation of all privacy guarantees
/settings	Yes	Permit management	Active permits, revocation

Appendix B — Glossary

Term	Definition
FHE	Fully Homomorphic Encryption — computes on encrypted data without decrypting
euint128	Encrypted 128-bit unsigned integer (position size, collateral, entry price)
euint64	Encrypted 64-bit unsigned integer (Pool 2 LP balances)
ebool	Encrypted boolean (long/short direction, liquidation flag)
Permit	EIP-712 signed authorisation granting decryption access to specific handles
Threshold Network	Fhenix's distributed key holders who cooperate to perform decryption
Handle	On-chain reference to an encrypted value (not the value itself)
FHERC20	Fhenix's encrypted ERC-20 standard (balances stored as euint64)
CoFHE	Co-Processor for FHE — Fhenix's off-chain compute layer
DP	Differentially Private — value has controlled noise added
ZK	Zero-Knowledge Proven — exact value with cryptographic correctness proof
Operator	Approved address that can call confidentialTransferFrom (Pool 2 setup)
Bucket	Price range containing position keys for liquidation discovery
FHEOS	FHE Operating System — Fhenix's off-chain FHE computation engine